Robustness of Retrieval-Augmented Language Models to Spurious Features: Literature Survey and Baseline Reproduction

NLP Research Project, Milestones M1 and M2

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***Abstract*—Retrieval-augmented language models (RALMs) answer questions from retrieved documents, but their answers can change when only the surface form of a document changes. We survey five papers on RAG robustness and select Yang et al. [1] as our anchor paper, which defines *spurious features* (style, source, logic, format and metadata) and measures sensitivity to them with the Robustness Rate (RR). We then reproduce the paper's baseline result for Llama-3.1-8B-Instruct on the released SIG_Wiki benchmark (Table 6 of [1]) on a free Kaggle T4 GPU, using 4-bit quantisation and greedy decoding. Our reproduced RR is much higher than reported for Style (56.5 vs. 10.0), Source (49.0 vs. 15.5), Logic (68.7 vs. 20.0) and Format (64.5 vs. 24.0), and close for Metadata (90.0 vs. 94.0). We analyse this gap and argue that it is mainly explained by how SIG_Wiki was selected (only pairs on which the authors' own runs were not robust), together with differences in decoding and precision. We also find that the RR metric treats "wrong before and after" as robust, which inflates the Metadata score, and that HTML formatting causes the largest drop in accuracy.**

***Index Terms*—retrieval-augmented generation, robustness, spurious features, reproducibility, large language models**

I. Introduction

Retrieval-augmented generation (RAG) gives a language model documents from a retriever and asks it to answer from them. This makes answers more factual, but only if the model reads the documents reliably. Earlier work studied *explicit* noise: retrieved documents that are irrelevant, distracting, false or placed in a bad position [2]–[5]. Yang et al. [1] study *implicit* noise: the document keeps exactly the same facts, and only its surface changes, for example plain text versus HTML.

This report covers the first two milestones of our project. Milestone M1 (Section II) surveys five papers, compares them, selects an anchor paper and checks its reproducibility. Milestone M2 (Sections III–V) reproduces one feasible baseline from the anchor paper and analyses the gap between our result and the reported one.

II. M1: Literature Survey

Research area: RAG Robustness and Reliability.

A. Selected Papers

P1, Yang et al., ACL 2026 [1] (anchor). The paper asks whether a RAG model's answer changes when the golden document keeps its meaning but changes its surface. It defines five types of spurious features: Style (simple or complex wording), Source (human-written, LLM-paraphrased or self-paraphrased), Logic (sentence order), Format (JSON, HTML, YAML, Markdown) and Metadata (timestamps and data-source URLs). The SURE framework injects these features and checks with bidirectional NLI and answer-string matching that the meaning is preserved. For each original/perturbed pair, $C = Y(y) - Y(\hat{y})$, where Y is answer correctness; the Robustness Rate (RR), Win Rate (WR) and Lose Rate (LR) are the shares of pairs with $C = 0$, $C = -1$ and $C = 1$. All tested models were sensitive, including GPT-4o (about 89% RR on the Twitter data-source perturbation). Larger models and Chain-of-Thought prompting did not remove the problem, while SFT and DPO on Llama-3.1-8B raised RR on SIG_Wiki above 94%.

P2, Cuconasu et al., SIGIR 2024 [2]. This paper studies which documents a retriever should place in the prompt, their position and their number. Using NQ-Open with a 21M-passage Wikipedia corpus and four 7B-class models in 4-bit, it finds that high-scoring but answer-free ("distracting") documents hurt accuracy, while adding random documents can improve accuracy by up to 35%.

P3, Yoran et al., ICLR 2024 [3]. This paper shows that irrelevant retrieved context can make a model answer wrongly. It tests an NLI filter, which prevents the drop but also removes useful passages, and fine-tuning Llama-2-13B with QLoRA on a mix of relevant and irrelevant contexts. As few as 1,000 training examples make the model robust to irrelevant context while keeping the benefit of relevant context.

P4, Chen et al., AAAI 2024 [4]. RGB is a bilingual (English and Chinese) benchmark of 600 base questions plus 200 integration and 200 counterfactual questions. It tests noise robustness, negative rejection, information integration

and counterfactual robustness on six LLMs, and finds that models handle some noise but struggle with rejecting unanswerable questions, combining documents and detecting false information.

P5, Liu et al., TACL 2024 [5]. This paper moves the answer-bearing document among 10–30 documents (2,655 NQ-Open queries) and also uses a synthetic key-value retrieval task. Accuracy follows a U-shaped curve: it is highest when the relevant information is at the start or end of the context and lowest in the middle.

B. Comparison

Table I compares the five papers. P2–P5 study explicit noise: what is retrieved, which abilities are missing, and where information is placed. P1 is the only one that keeps the content fixed and changes only the surface of the document.

C. Anchor Paper Selection

We select P1 [1] for four reasons. *Scientifically meaningful:* it introduces a kind of noise (implicit, meaning-preserving) that P2–P5 do not measure, and an instance-level metric that catches answer changes hidden by average accuracy. *Partially reproducible:* the authors release code and the SIG_Wiki benchmark of ready-made original/perturbed pairs, so evaluation does not require regenerating perturbations with a 70B model. *Computationally feasible:* evaluating an 8B model on SIG_Wiki is about 3,000 short generations, which fits a free T4 GPU with 4-bit loading. *Extensible:* its limitations (one golden document, sampled decoding, expensive mitigation) each suggest a one-variable experiment. P2 is the easiest to reproduce, but its research question is older and has been widely revisited; P3 and P5 need larger models or long contexts; P4 depends partly on ChatGPT.

D. Reproducibility Checklist

Table II applies the guideline's checklist to the anchor paper, and Table III summarises it for all five papers.

E. Initial Research Problem

P1 measures robustness with one golden document per prompt, sampled decoding (temperature 0.1), and fixes it with full fine-tuning on eight A100 GPUs. This leaves two questions that can be studied on a free GPU: (1) how much of the measured non-robustness is run-to-run instability rather than a real effect of the perturbation, and (2) whether a training-free step, such as converting HTML, JSON or YAML back to plain text before answering, reduces sensitivity to format and metadata features. Our M2 results (Section V) sharpen these questions for M3.

III. M2: Baseline Reproduction Setup

A. What We Reproduce

We reproduce the row "Llama3.1-8B (Wiki)" of Table 6 in [1]: the Robustness Rate of Llama-3.1-8B-Instruct [6] on SIG_Wiki, per spurious-feature category, before any SFT/DPO. We do not reproduce building SURE_Wiki (it needs Llama-3.1-70B for perturbations) or the SFT/DPO results (they need 8×A100). Table IV lists the setup.

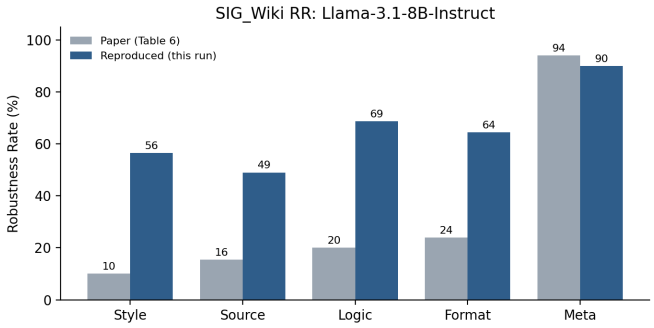


Fig. 1. Reported (Table 6 of [1]) vs. reproduced RR per category.

B. Prediction Before Running

Before running the experiment we predicted that our RR would be *higher* than the reported values for Style, Source, Logic and Format, because SIG_Wiki contains only pairs on which the authors' models were not robust in their own runs, and a fresh run should regress towards the model's typical behaviour. We predicted Metadata would stay high (≥90%).

IV. M2: Results

Table V and Fig. 1 compare our RR with the paper. Our RR is higher by 33.5 to 48.7 points for Style, Source, Logic and Format, and 4.0 points lower for Metadata. The prediction holds for four of five categories, and Metadata stayed above 90% as predicted. Over all 1,500 pairs, correctness was unchanged for 69.0% of pairs, accuracy was 49.9% on original documents and 47.2% on perturbed ones, and WR (14.1%) and LR (16.9%) were similar.

Table VI gives the per-perturbation results. HTML is the most harmful perturbation: accuracy falls from 69% to 28% and LR is 47%. Markdown is the opposite: accuracy rises from 51% to 65%, with WR 17% and LR only 3%.

V. M2: Reproduction Gap Analysis

We ran one configuration, so we cannot separate the factors below experimentally; for each we state the expected direction of its effect and the evidence we have.

Split (selection effect). SIG_Wiki was built by keeping, for each perturbation, 100 pairs "where both models lack robustness" [1] in the authors' runs. Such a set is selected on the outcome, so a new run on the same pairs is expected to show more robustness (regression to the mean). This predicts a large positive gap for the four filtered categories, which is what we observe. Evidence that many flips are unstable rather than systematic: over all pairs WR (14.1%) is close to LR (16.9%), i.e. perturbations make the model right almost as often as wrong. HTML is the clear exception (LR 47% vs. WR 6%). The paper does not state whether Metadata pairs were filtered the same way; its reported 94% RR suggests they were not, which fits our small gap there. We consider this the main cause of the gap.

Seed and decoding. The paper samples at temperature 0.1; we decode greedily. Sampling adds run-to-run variation, so

TABLE I
COMPARISON OF THE FIVE SURVEYED PAPERS

	P1 Yang et al. [1] (anchor)	**P2 Cuconasu et al. [2]**	**P3 Yoran et al. [3]**	**P4 Chen et al. [4]**	**P5 Liu et al. [5]**
Research problem	Do meaning-preserving surface changes alter RAG answers?	Which documents, positions and counts help RAG?	Irrelevant context makes RALMs answer wrongly	Which RAG abilities do LLMs lack?	Do LMs use the middle of long contexts?
Method	SURE: 5-type taxonomy, feature injection, NLI preservation check, RR/WR/LR; SFT and DPO	Controlled prompts mixing gold, distracting and random documents	NLI filter; QLoRA fine-tuning on mixed relevant/irrelevant contexts	RGB benchmark with four testbeds	Vary position of the answer document; key-value retrieval
Dataset	NQ-Open + Wikipedia (SURE_Wiki, SIG_Wiki); TriviaQA + Bing (SIG_Trivial)	NQ-Open, 21M passages, 2,889 test queries	NQ, 2WikiMQA, StrategyQA, Bamboogle, Fermi	RGB: 600 + 200 + 200 questions, EN/ZH	NQ-Open (2,655 queries); synthetic key-value
Models	Mistral-7B, Llama-3.1-8B; GPT-4o, Qwen2.5, DeepSeek	Llama2-7B, Falcon-7B, Phi-2, MPT-7B (4-bit)	Llama-2-13B	ChatGPT, ChatGLM(2)-6B, Vicuna-7B, Qwen-7B, BELLE-7B	GPT-3.5-Turbo, Claude-1.3, MPT-30B, LongChat-13B
Metric	RR, WR, LR	Accuracy	Accuracy	Accuracy, rejection, error detection/correction	Accuracy
Code / data	Yes (SIG benchmarks only)	Yes	Yes	Yes	Yes
Main finding	All models sensitive; scale and CoT do not fix it; SFT/DPO do	Distracting docs hurt; random docs can help (+35%)	1k examples give robustness to irrelevant context	Weak at rejection, integration, counterfactuals	U-shaped accuracy vs. position
Limitation	Single golden document; 8×A100 for training; taxonomy may be incomplete	Why random noise helps is unexplained; small models	Needs training; live web search not repeatable	Older models; news domain; ChatGPT versions change	Older models; position only, not style or format
Possible extension	Noise floor of RR; training-free mitigation	Spurious features inside distractors	Does it also give spurious-feature robustness?	Add a spurious-feature testbed	Interaction of format and position

TABLE II
REPRODUCIBILITY CHECKLIST FOR THE ANCHOR PAPER (P1)

Question	**Answer**	**Evidence**
Public dataset?	Partially	SIG_Wiki (16 perturbations × 100 pairs) is in the repo; full SURE_Wiki is not
Dataset size documented?	Yes	SURE_Wiki sizes per category; 100 pairs per perturbation in SIG_Wiki
Public code?	Yes	github.com/maybenotime/RAG-SpuriousFeatures
Pretrained model?	Yes	Llama-3.1-8B-Instruct, Mistral-7B-Instruct-v0.3 (gated on Hugging Face)
Hardware known?	Yes	A100 80GB with vLLM; training on 8×A100
Training time known?	No	Epochs and learning rates given, wall-clock time not reported
Dependencies listed?	Partially	No README or requirements file; imports show vLLM, transformers, faiss, DeepSpeed
Metric defined?	Yes	RR/WR/LR formula; scoring code public
Feasible on free GPU?	Partially	Evaluation yes (4-bit, T4); perturbation generation and SFT/DPO no
Results reproducible?	Partially	Temperature-0.1 sampling, hard-coded paths in scripts, and SIG_Wiki selected from the authors' own runs

TABLE III
REPRODUCIBILITY SUMMARY FOR ALL FIVE PAPERS

Paper	**Code**	**Data**	**Models**	**Free GPU?**
P1 [1]	Yes	Partial	Yes (gated)	Evaluation only
P2 [2]	Yes	Yes	Yes	Partly (21M-doc index)
P3 [3]	Yes	Yes	Yes (gated)	Partly (13B, paid search)
P4 [4]	Yes	Yes	Partial	Yes (open models)
P5 [5]	Yes	Yes	Partial	Partly (long context)

a sampled run flips more pairs and gives lower RR than a deterministic run. This also lowers the RR of the run that was used to select SIG_Wiki. Expected direction: our RR higher. We did not measure seed variance in this milestone.

Preprocessing. We copied the authors' document construction exactly. One difference is the chat template: we add the assistant generation prompt, which the authors' generic script does not. This makes the model answer in its normal assistant format. Expected effect: small, direction unknown.

Implementation. We use Hugging Face `generate` with left padding and a 64-token limit instead of vLLM. Answers are short (median 3 words, mean 7.9, max 58), so the token limit rarely matters. Expected effect: small.

Model version and precision. We use the same Llama-3.1-

TABLE IV
REPRODUCTION SETUP: PAPER VS. OURS

Item	Paper [1]	Ours
Dataset	SIG_Wiki, 16 perturbations × 100 pairs	Same file from the authors' repo; 1,500 pairs (15 perturbations; only the self-paraphrase subset of the evaluated model)
Split	Test-only benchmark	Same; no split
Preprocessing	Authors' prompt construction	Same: original = title + text (HTML page for metadata); perturbed document as released
Model	Llama-3.1-8B-Instruct, bf16	Same weights, 4-bit NF4 [7]
Prompt	Extract answer (≤5 tokens) or reply NO-RES	Same instruction and layout; chat template with generation prompt
Decoding	Temperature 0.1 (vLLM)	Greedy; max 64 new tokens; batch size 8 (HF Transformers)
Training	None for this row	None
Metric	RR/WR/LR, normalised answer-in-prediction	Same, using the authors' `normalize_answer` code; category RR = mean over its perturbations
Hardware	NVIDIA A100 80GB	Kaggle, 2× Tesla T4 (16GB)
Run time	Not reported	34 min 49 s (whole notebook)
Seeds	Not reported	One deterministic greedy run

TABLE V
ROBUSTNESS RATE (%) ON SIG_WIKI, LLAMA-3.1-8B-INSTRUCT

Category	Paper RR	Our RR	Gap	Our WR	Our LR
Style	10.0	56.5	+46.5	21.5	22.0
Source	15.5	49.0	+33.5	27.5	23.5
Logic	20.0	68.7	+48.7	13.3	18.0
Format	24.0	64.5	+40.5	13.3	22.3
Meta	94.0	90.0	−4.0	5.3	4.8

TABLE VI
PER-PERTURBATION RESULTS (%), 100 PAIRS EACH

Type	Perturbation	RR	WR	LR	Acc_o	Acc_p
Style	Simple	53	19	28	60	51
	Complex	60	24	16	45	53
Source	LLM-generated	50	27	23	51	55
	Self-generated	48	28	24	51	55
Logic	Reverse	68	13	19	57	51
	Random	66	15	19	57	53
	LLM-reranked	72	12	16	60	56
Format	HTML	47	6	47	69	28
	JSON	65	12	23	63	52
	YAML	66	18	16	54	56
	Markdown	80	17	3	51	65
Meta	Timestamp (pre)	94	0	6	40	34
	Timestamp (post)	89	4	7	34	31
	Source: Wiki	88	12	0	28	40
	Source: Twitter	89	5	6	29	28

8B-Instruct weights but quantised to 4-bit NF4. Quantisation can change borderline answers in either direction. It may also explain why the model often refuses: it answers NO-RES on 28.1% of original documents even though every golden document contains the answer, which keeps accuracy near 50%. Expected effect: unknown direction; it lowers accuracy rather than RR directly.

Compute limits. A T4 has 16GB of memory, so 4-bit loading was required and bf16 inference was not possible. We could not reproduce perturbation generation or the SFT/DPO rows, and we used one run instead of several seeds.

Metric observation. RR counts a pair as robust when the answer is wrong both before and after the perturbation. In the Metadata category, 62.0% of pairs are wrong both times (mostly NO-RES on the HTML-page original), and only 28.0% are correct both times. The 90% Metadata RR therefore says more about consistently failing than about being robust. Over all pairs, 35.9% are wrong both times.

TABLE VII
EXAMPLES WHERE THE PERTURBATION BROKE A CORRECT ANSWER

Pert.	Question (gold)	Original	Perturbed
HTML	Who wrote the music of Phantom of the Opera? (Andrew Lloyd Webber)	Andrew Lloyd Webber	NO-RES
Twitter source	Who has the power to approve or veto legislation? (the President)	the president	NO-RES
Simple style	What year did The Price Is Right first air? (1956)	…first aired in 1956.	1972
Random order	Glycogen and amylopectin are chains of which sugar? (glucose)	glucose	hexoses
Self-generated	Salisbury steak vs. hamburger steak difference? (ingredients)	ingredients	composition

A. Representative Failures

Table VII shows pairs where the original answer was correct and the perturbed one was not. Three patterns appear: (1) refusal: format and data-source changes make the model reply NO-RES although the answer is still in the document (105 of 253 lose cases); (2) wrong span: simplified or shuffled text leads to a different, wrong entity; (3) strict scoring: correct but paraphrased answers ("composition" for "ingredients") are marked wrong by string matching. Scoring can also be lenient: in one Markdown "win" the model copied a passage that contained a gold cast name and then replied NO-RES, yet string matching marked it correct.

VI. CONCLUSION AND NEXT STEPS

We surveyed five RAG robustness papers and chose Yang et al. [1] as the anchor because it studies a new, implicit kind of noise and its evaluation can be reproduced on a free GPU. Reproducing its Llama-3.1-8B SIG_Wiki baseline gave much higher RR than reported for four of five categories.

The gap is best explained by the outcome-based selection of SIG_Wiki, with further contributions from greedy decoding and 4-bit precision; it is evidence about the benchmark, not a failed reproduction. For M3 we will test one variable at a time. Candidate hypotheses are: (a) the run-to-run flip rate with *no* perturbation explains a substantial share of the reported non-robustness, measured with several sampling seeds; and (b) converting HTML, JSON and YAML documents to plain text before reading reduces the Format lose rate, especially the HTML lose rate of 47%.

Code and outputs: https://github.com/⟨your-username⟩/rag-robustness-reproduction

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